Introduction & Hook

"When you press a button in VR and a door opens, how does the system know what to do?"

Explain:

VR environments are interactive because of **scripts** that respond to **user events** such as:

- Controller input
- Hand gestures
- Voice commands
- Object collisions

These scripts define the behavior of objects.

PART 1: What are Scripts in VR?

Definition of Scripts

A **script** is a small program that controls the behavior of objects in a virtual environment.

Scripts can:

- Detect user input
- Trigger actions
- Control animations
- Manage interactions between objects

Common scripting languages in VR:

- **C#** (Unity)
- **C++ / Blueprints** (Unreal Engine)
- **JavaScript** (WebXR)

Example

In a VR training simulation:

User presses button → machine starts operating.

This behavior is written in a script.

PART 2: Event-Driven Scripts

What is Event-Driven Programming?

Definition:

Event-driven programming is a programming approach where actions occur in response to specific events.

An **event** is something that happens during the program.

Examples of events:

- Button pressed
- Controller trigger pulled
- Object collision
- Hand gesture detected

Event-Driven Structure

Basic structure:

```
Event → Script executes → Action performed
```

Example:

```
If controller trigger pressed
    Open virtual door
```

Example in VR Game

Event: User touches object
Script: Detect collision
Action: Object changes color

Common Events in VR

- Controller input
- Head movement
- Object interaction
- Voice command
- Physics collision

🎮 PART 3: Interaction Techniques

Simple Interaction Techniques

Simple interactions involve **basic user actions**.

Examples:

- Press button to open door
- Grab an object
- Select menu option
- Teleport to location

These interactions usually involve:

1 user input → 1 action

Example:

User presses trigger → object moves.

Complex Interaction Techniques

Complex interactions involve **multiple inputs or system behaviors**.

Examples:

- Two-hand object manipulation
- Gesture recognition
- Physics-based object movement
- Multiplayer interactions

Example:

User grabs a virtual tool → rotates it → activates machine.

This requires:

- Tracking hand movement
- Detecting collisions
- Running multiple scripts simultaneously

Interaction Loop in VR

```
User Action
   ↓
Input Detection
   ↓
Script Execution
   ↓
Environment Response
```

Real-World VR Examples

Examples of interaction techniques:

VR Training

User presses control panel → machine starts.

VR Medical Simulation

User picks surgical tool → performs operation.

VR Gaming

Player grabs weapon → shoots target.

All of these depend on scripts responding to events.

Recap & Questions

Key Points

- Scripts control object behavior in VR
- Event-driven programming triggers actions based on user input
- Simple interactions involve single actions
- Complex interactions involve multiple events and behaviors

Quick Questions

1. What is a script in VR?
2. What is event-driven programming?
3. Give an example of simple interaction.
4. Why are complex interactions harder to implement?